STRATEGIC SOURCING & PROCUREMENT

The Final Chapter:
Quality You Can Prove

Implementing & Validating Quality Management Systems

💬 Opening question: Can you name a company whose quality failure made global headlines — and what went wrong?

KBS · KFUPM | MSc Supply Chain Management

SETTING THE STAGE

What This Course Has Built — And How Today Completes It

✓ Quality Perspectives	✓ Quality Tools: QFD, SERVQUAL, Flow Charts,….	✓ Quality Teams and Project Management
✓ Quality standards and Evaluation	✓ Quality Control Methods: Variables and Attributes	→ Implementing & Validating Quality Systems

Today we answer: How do you IMPLEMENT quality — and how do you PROVE it is working?

TODAY'S AGENDA

Six Themes — One Final Lecture

01
The Quality System Model
Three spheres · building blocks

02
Three Spheres Deep Dive
Planning · Assurance · Control

03
Assessing Your System
Internal analysis · self-assessment

04
Quality Audits
Types · process · supplier audits

05
Case Studies
Toyota · Saudi Aramco · Boeing

06
Simulation + Discussion
You run the audit

07
Validation Playbook
8-step process to prove quality

08
Digital & AI Frontier
QMS 4.0 · your future career

SECTION 1 — QUALITY SYSTEM MODEL

What Is a Quality System? More Than Inspection.

A quality system is the entire organisational architecture that ensures products and services consistently meet customer requirements — verifiably and at minimum cost.

INTEGRATIVE APPROACH · Strategic Alignment

Enterprise Capabilities · Three Spheres of Quality

Information & Finance · Closeness to Customers

Culture · Organisational Learning

PEOPLE — The Core

BUILDING BLOCKS

Six Building Blocks — One Interconnected System

People

Intellect, **empathy & motivation** — the core of all capability

Org. Learning

Lifelong learning · training design · knowledge as capital

Culture

Trust, openness, attitude toward change — defines what gets done

Customer Closeness

Understand changing needs · retained customers = profitability

Information & Finance

IT systems + financial resources = infrastructure for service

Integrative Approach

Cross-functional glue — quality is everyone's responsibility

The Three Spheres: Planning → Assurance → Control

Every quality action in your organisation sits inside one of three spheres. Miss any sphere and your system has a blind spot.

Quality Planning & Management

- Set quality objectives aligned to strategy
- Design processes to meet requirements
- Define quality standards before production starts
- Resource allocation for quality activities

Quality Assurance

- Systematic activities providing confidence
- Process audits & documentation reviews
- ISO 9001 certification activities
- Supplier quality agreements & audits

Quality Control

- Detect and correct defects in outputs
- SPC charts · inspection · sampling
- Conformance measurement vs. spec
- Corrective action on non-conformances

💡 Ask yourself: Does your supplier have all three spheres? Or just quality control?

Internal Validation: How Do You Know Quality Is Working?

Before you can improve, you must measure. Internal validation uses three linked tools:

STEP 1	**Environmental Analysis** Identify internal strengths and gaps against quality objectives. Ask: Where are we strong? Where are we exposed?
STEP 2	**Self-Assessment Tool** Score your organisation across all quality building blocks. Rate each area 1–5. Gap = target minus current score.
STEP 3	**Quality Audit** Independent, structured review of whether the quality system is functioning as documented. The proof, not just the score.

🎯 **Self-assessment is not self-congratulation. Honest gaps drive genuine improvement.**

AUDIT TYPES

Not All Audits Are Equal — Know Which One You Need

Operational Audit

Reviews all management functions — helps managers discharge responsibilities effectively

Performance Audit

Measures outcomes against set targets — are you achieving what you said you would?

Supplier Audit

Evaluates a supplier's quality system before or during the relationship — critical for procurement

Certification Audit

External body verifies conformance to ISO 9001, ISO 14001, etc. — issued or denied

Award Audit

For quality awards (MBNQA, EFQM, King Abdulaziz Quality Award) — benchmarking tool

Presidential Audit

CEO-led audit — signals top-level commitment to quality culture

Qualitative Audit

Compares current practice against structural and behavioural benchmarks. Harder to quantify but critical for culture.

Quantitative Audit

Conformance rates, cost-of-quality data, control chart analysis, SPC. Numbers that prove (or disprove) quality.

The Supplier Audit: Your Most Powerful Procurement Tool

A supplier audit is not just quality inspection — it is the mechanism by which procurement professionals validate that their supply base is a competitive asset, not a liability.

PRE-AUDIT

- Define scope: product lines, processes, facilities
- Share audit criteria 30 days in advance (ISO 19011)
- Assign auditor with no conflict of interest
- Build supplier-specific checklist

DURING AUDIT

- Opening meeting: confirm scope and timeline
- Observe processes against documented procedures
- Interview personnel at all levels
- Collect objective evidence — photos, records, samples

POST-AUDIT

- Exit meeting: share findings, no surprises
- Issue formal report within 10 business days
- Require Corrective Action Plans (CAPs) for all findings
- Track CAP implementation — no close-out without evidence

📌 Key principle: A supplier audit finding is a gift — it tells you what to fix before it reaches your customer.

SECTION 5 — CASE STUDIES

Three Companies. Three Quality Lessons. All Real.

Quality system failure is not abstract. These cases show what happens when the system breaks — and how recovery works.

Toyota

Manufacturing | Japan / Global

Accelerator pedal recall — 9 million vehicles. What happens when quality control catches a defect too late.

Saudi Aramco

Energy | Saudi Arabia

IKTVA quality programme transformation — how a national energy company built world-class supplier quality validation.

Boeing 737 MAX

Aerospace | USA / Global

Two crashes, 346 lives. The catastrophic cost of separating quality assurance from engineering sign-off.

CASE STUDY 1 — TOYOTA

Toyota 2009–2010: When Quality Control Was Not Enough

Toyota — the company that invented lean and the Toyota Production System — faced the largest automotive recall in history. How?

9M+	$1.2B	37	−16%
Vehicles recalled globally	US federal fine paid	Deaths linked to defects	Share price drop 2010

What Failed

Suppliers pressured to cut costs — material substitutions made without full quality sign-off. QC caught the problem only at the customer.

The Root Cause

Quality planning did not include early supplier involvement. The three-sphere model was broken — assurance was weak in the supply base.

The Recovery

Toyota implemented Global Quality Task Force. New supplier auditing standards. Top-management quality reviews quarterly. Cost: $2B+ in 2 years.

The Lesson for You

Quality control at the end of the line is too late and too expensive. Assurance and planning upstream is where procurement earns its value.

CASE STUDY 2 — SAUDI ARAMCO

Saudi Aramco: Building Quality Into the Supply Nation

With SAR 200B+ annual procurement, Aramco transformed supplier quality from a compliance exercise into a national capability-building programme.

SA

IKTVA Quality Mandate

In-Kingdom Total Value Add programme requires suppliers to meet defined quality standards — not just price competitiveness. Quality is a qualifying threshold.

Supplier Quality Index (SQI)

Aramco scores every major supplier quarterly: delivery performance, defect rates, audit findings, corrective action closure speed. Below threshold = improvement plan or delisting.

First Article Inspection

Before any new product is accepted into the supply chain, it must pass First Article Inspection — documented proof that the manufacturing process can produce to specification.

Supplier Development Programme

Aramco invests in training its suppliers' quality teams — ISO 9001 implementation, SPC training, 8D problem solving. Quality capability is co-built, not just demanded.

Vision 2030 context: Supplier quality capability = national industrial capability. Quality is not just good practice — it is national strategy.

CASE STUDY 3 — BOEING

Boeing 737 MAX: What Happens When Assurance Fails

346 people died in two crashes (2018–2019). The cause: a software system (MCAS) approved without adequate quality assurance review — and a culture where safety concerns were not escalated.

Control Failure

Pilot error warnings from simulator testing were flagged and then deprioritised under schedule pressure. Quality control data was available — it was ignored.

Assurance Failure

FAA delegated safety certification to Boeing engineers reporting to Boeing management. No independent quality assurance on the critical MCAS sign-off.

Cost of Failure

$20B+ total cost including 20-month grounding, litigation, compensation, reputational damage. All three CEOs during the crisis lost their roles.

The Lesson

The three spheres — Planning, Assurance, Control — must all function independently. When any sphere is captured by production pressure, people die.

SECTION 6 — SIMULATION EXERCISE

Your Turn: You Are the Quality Manager

This is a decision-making exercise — not a quiz. You will face the choices real quality and procurement professionals make under pressure.

01

Your Role

You are QMS Manager at AlWatani Packaging — 500-employee manufacturer, Dammam Industrial City

02

The Scenario

ISO 9001 surveillance audit in 60 days. Certification body arrives. Three open non-conformances from last year.

03

The Problem

Your largest customer just reported a 4.2% defect rate on last month's shipment — above the 1.5% contractual limit

04

The Reveal

We compare team decisions and audit readiness scores after 15 minutes of team discussion

⏱ minutes | Teams of 4 | One decision sheet per team

SIMULATION — SCENARIO BRIEF

AlWatani Packaging: The 60-Day Audit Crisis

lWatani Packaging, Dammam | ISO 9001:2015 Certified | SAR 180M revenue | Key customer: SABIC Packaging Division

Surveillance audit in 60 days. Last audit: 3 minor non-conformances (NCs) — none closed. Customer defect complaint received this week.

Open Non-Conformance #1

Calibration records for 3 inspection gauges are 8 months overdue. Required: annual. Evidence of use: daily.

Open Non-Conformance #2

Supplier qualification records for 2 raw material suppliers are not on file. Suppliers have been used for 18 months.

Open Non-Conformance #3

Management review meeting not conducted in 14 months. ISO 9001 requires at minimum annual frequency.

Customer Complaint

SABIC reports 4.2% defect rate (contractual limit: 1.5%) on three shipments. 8D report required within 5 business days.

SIMULATION — YOUR DECISIONS

Five Decisions — 15 Minutes — As a Team

Decision 1: NC Prioritisation	Which of the 3 NCs is most critical to close first before the audit — and why? What is your closure strategy and evidence needed?
Decision 2: Customer Complaint Response	Draft the structure of your 8D report. What are steps D1–D3 (Team, Problem Description, Containment)? What containment action do you implement today?
Decision 3: Management Review	The management review is 14 months overdue. You have 60 days. What is on the agenda? Who must attend? What records do you produce?
Decision 4: Audit Preparation	List your top 5 preparations for the surveillance audit. What is the single biggest risk to your certification status?
Decision 5: Future Prevention	After the audit, what one structural change do you make to your quality system to prevent this situation from recurring next year?

📝 **Record your team's decisions. Debrief in 15 minutes — we score each team's audit readiness.**

SIMULATION — DEBRIEF

What the Best-Performing QMS Teams Do Differently

Companies that pass surveillance audits and maintain low customer defect rates consistently do these six things:

01 NC Management as Routine

They treat corrective action closure as a KPI tracked weekly — not something done before audits.

02 Management Review is Business Review

Quality data is part of the executive dashboard. Leaders do not see quality reports only at audit time.

03 Customer Complaint = Learning Event

Every customer complaint triggers a formal 8D process — not just an apology email. Root cause is mandatory.

04 Calibration & Supplier Records

Master calibration schedules in the QMS software with automated alerts. No manual chasing.

05 Internal Audits Ahead of Surveillance

Internal audit programme covers all processes annually. Surveillance audit findings are never a surprise.

06 Documented Signatures ≠ Quality

Records are evidence of the system — but they are not the system. Culture and capability are.

OPEN DISCUSSION — 1

Discussion: Quality in the Saudi Context

Three questions — 3 minutes each. Discuss with your neighbour. We hear two responses per question.

Q1 Saudi Arabia has the King Abdulaziz Quality Award. How does it compare to the Malcolm Baldrige Award in the US? What does winning such an award actually prove — and what doesn't it prove?

Q2 SABIC, NEOM and Vision 2030 industrial projects all require supplier quality certification. But many Saudi SMEs are not ISO 9001 certified. Is certification a barrier to growth — or a pathway?

Q3 You are the CPO of a Saudi manufacturer. Your quality audit reveals your top supplier has passed ISO 9001 for 3 years but your defect rate from them is rising. What has gone wrong — and what do you do?

No right answer — only honest thinking. That is what quality management requires.

SECTION 7 — VALIDATION PLAYBOOK

Validating Your Quality System: The 8-Step Process

Validation is the annual proof that your quality system is not just documented — but effective. Here is how to do it.

1 Assemble the Right Team

QM + Operations + Procurement + top management sponsor

2 Review Your Results

KPIs, customer complaints, internal audit findings, supplier scorecards

3 Identify What Is Working

Document strengths — which processes consistently meet targets?

4 Identify What Is Not

Gap analysis — where is the system underperforming against objectives?

5 Identify Unaddressed Areas

Aspects of quality deployment not yet covered — emerging risks, new processes

6 Develop Your Signature Strategy

One or two major quality initiatives that define this year's focus

7 Build the Implementation Plan

Actions · owners · timescales · resource requirements · costs

8 Repeat Annually

Quality is a cycle — last year's improvement becomes this year's baseline

OPEN DISCUSSION — 2

Discussion: You Are the Quality Leader — What Do You Do?

You are the VP Quality of a mid-size Saudi FMCG manufacturer. You have SAR 1.5M and 12 months to transform your quality system.

Option A

Invest in ISO 9001:2015 certification for your two key suppliers who are currently uncertified. External audit, training, 12-month implementation.

Option B

Deploy a digital QMS platform (SAP QM or MasterControl). Digitise all quality records, automate audit scheduling, enable real-time KPI dashboards.

Option C

Launch a company-wide quality culture programme. 8D training for all engineers, SPC training for all production teams, monthly quality town halls with CEO.

Option D

Hire 3 experienced quality engineers, restructure the QA team, and implement a dedicated supplier development programme for your top 10 suppliers.

🗳 **Vote: Choose your option. Then: what ONE metric tells you it worked after 12 months?**

QMS 4.0: Quality Management in the Digital Age

This is your competitive edge as the next generation of supply chain and quality leaders.
The tools exist. The question is whether you will lead their adoption.

AI-Powered Audit Systems

Digital QMS Platforms

Predictive Quality Analytics

Blockchain Traceability

IoT / Smart Inspection

ESG Quality Compliance

DIGITAL — QMS PLATFORMS & TOOLS

The Modern QMS Tech Stack: What Real Companies Use

Know these tools — they are in procurement and quality job descriptions across Saudi Arabia and the GCC.

Enterprise QMS

SAP QM · Oracle Quality · IFS Quality

End-to-end: document control → NCRs → CAPA → audit → compliance. Integrated with ERP.

Standalone QMS

MasterControl · ETQ Reliance · Qualio · Veeva QualityDocs

Cloud-based. Document control, training records, audit management, CAPA tracking.

Supplier Quality

Siemens Opcenter Quality · Intelex · Sparta Systems

Supplier scorecard management, First Article Inspection, supplier corrective actions.

AI & Analytics

Palantir Foundry · C3.ai Quality · Sight Machine

Predictive defect detection using ML. Identify failure modes before they produce rejects.

Inspection & IoT

Cognex Vision Systems · Keyence · Landing AI

Automated visual inspection replacing manual. 100% inspection at machine speed.

Saudi / GCC Specific

Etimad · SFDA eProcurement · SASO eServices

Government quality and conformity portals mandatory for Saudi market access.

DIGITAL — AI & PREDICTIVE QUALITY

AI in Quality Management: From Reactive to Predictive

Traditional QMS: defect detected → corrective action. AI-enabled QMS: defect predicted → prevented before it occurs.

📈 Predictive SPC
ML models detect process drift 2–4 hours before defects appear. Used by Bosch, ABB in Saudi facilities. Reduces scrap by 20–35%.

👁 Automated Visual Inspection
Computer vision checks 100% of parts at production speed. No sampling error. Toyota uses Cognex systems for zero-defect stamping lines.

⚠️ Supplier Risk AI
Algorithms monitor 10,000+ supplier signals: financial health, news, shipment delays, quality history. Early warning 6–8 weeks ahead.

📅 AI Audit Scheduling
AI analyses which processes are highest risk and schedules internal audits dynamically — not on a fixed calendar that auditors learn to game.

📄 Natural Language Quality Reports
LLMs (GPT-4 class) read customer complaints, NCRs, and audit findings in seconds and cluster root causes automatically.

👁‍🗨 Digital Twin Quality
Virtual replica of your production line. Simulate the impact of a process change on quality output before physical implementation.

📌 **Platforms: SAP QM + AI, C3.ai, MasterControl AI, Siemens Xcelerator — all available to Saudi manufacturers today.**

ISO 9001:2015 in the Digital Era: What Changed

ISO 9001:2015 was designed for the digital age. Its risk-based thinking and context analysis require data — and digital QMS tools deliver it.

Risk-Based Thinking (Cl. 6)

Digital QMS automates risk assessment scoring and links risks to processes, controls, and audit findings.

Context of the Organisation (Cl. 4)

Stakeholder analysis tools and strategic issue tracking digitally maintained — no more static Word documents.

Performance Evaluation (Cl. 9)

Real-time KPI dashboards replace quarterly paper reports. Management reviews informed by live data.

Documented Information (Cl. 7.5)

Cloud QMS eliminates version control chaos. One controlled document. Accessible anywhere. Audit trail automatic.

🎯 **Saudi SASO and ZATCA are moving to digital conformity verification. Quality professionals who cannot operate digital QMS systems will struggle in the 2026+ market.**

The Quality Professional of 2030: What You Must Be

Quality management is transforming faster than any other supply chain function. The skills gap creates opportunity for those who prepare.

Data & Analytics

Read control charts, SPC data, quality dashboards. Interpret trends before they become crises.

Digital QMS Fluency

Know SAP QM, MasterControl, or equivalent. Be the person who trains others — not who avoids the system.

Supplier Quality Management

Supplier audits, corrective action management, APQP/PPAP — critical skills for procurement-quality interface.

Risk-Based Thinking

ISO 9001:2015 is risk-based. Identify, assess, and control quality risk — proactively, not reactively.

ESG & Sustainability Quality

Carbon footprint of quality failures. Scope 3 quality requirements. Green procurement conformance standards.

Arabic + English + Data

Bilingual quality professionals who speak data are rare and extremely valuable in the Saudi industrial market.

COURSE CONCLUSION

What This Course Has Built in You

Strategic Sourcing & Procurement is not about buying at the lowest price.
It is about creating a supply base that is a genuine competitive advantage.

1

Think System

Quality is not a department — it is an architecture of people, culture, information, and processes.

2

Plan → Assure → Control

All three spheres must function. If any is missing, your system has a dangerous blind spot.

3

Audit as Intelligence

Audits are not compliance exercises — they are the most reliable source of ground-truth about your supply chain.

4

Validate Annually

The quality system is only real if it is validated. Measure, gap-analyse, improve, repeat.

5

Embrace Digital

QMS 4.0 tools exist and are already in your competitors. Lead their adoption — do not resist it.

Thank You

"Quality is never an accident; it is always the result of intelligent effort."
— John Ruskin

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Three questions to carry into your career:

- Does your organisation have all three spheres of quality — or just quality control?
- When did you last see quality evidence (not just quality documentation)?
- Which digital quality tool will you champion in your first role — and how?

Good luck. Make quality a habit, not a compliance exercise.